
Software Requirements Specification

for

AI Connect

Version 1.0 approved

Prepared by Justin Whittington, Kenneth Mack, William Davis

Mississippi State University

10/6/2024

Table of Contents

Table of Contents	ii
Revision History	ii
1. Introduction.....	1
1.1 Purpose.....	1
1.2 Document Conventions.....	1
1.3 Intended Audience and Reading Suggestions	1
1.4 Product Scope	1
1.5 References.....	1
2. Overall Description	2
2.1 Product Perspective.....	2
2.2 Product Functions	2
2.3 User Classes and Characteristics	2
2.4 Operating Environment.....	2
2.5 Design and Implementation Constraints	2
2.6 User Documentation	2
2.7 Assumptions and Dependencies	3
3. External Interface Requirements	3
3.1 User Interfaces	3
3.2 Hardware Interfaces	3
3.3 Software Interfaces	3
3.4 Communications Interfaces	3
4. System Features	3
4.1 System Feature 1.....	Error! Bookmark not defined.
4.2 System Feature 2 (and so on).....	4
5. Other Nonfunctional Requirements	9
5.1 Performance Requirements	9
5.2 Safety Requirements	10
5.3 Security Requirements	10
5.4 Software Quality Attributes	10
5.5 Business Rules	10
6. Other Requirements	10
Appendix A: Glossary.....	10

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This SRS document will provide an outline of the software requirements for the web application, "AI Connect". It will provide a detailed product description as well as a framework to be used by the team.

1.2 Document Conventions

This document is divided into 6 major sections with subsections under each major section. Each section is denoted by a primary header with a number and the focus of that section. Each subsection is denoted by a secondary header with a sequential decimal and the focus of that subsection.

1.3 Intended Audience and Reading Suggestions

This document is written for the developers of AI Connect, as well as the client, and any faculty overseeing the project. Section 2 of this document will contain a general description of the project and how it should be used. Section 3 of this document will contain information of the different interfaces of the project. Section 4 of this document will contain functional requirements and features of the project. Section 5 of this document will contain information on non-functional requirements. If necessary, section 6 of this document will contain other requirements not contained in either section 4 or 5.

1.4 Product Scope

The project will be a web application used by academia and industry to propose and request seminars for a professor to provide to their class.

1.5 References

<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location.>

2. Overall Description

2.1 Product Perspective

This project was proposed by Charan Gudla as a senior project for software engineering majors at Mississippi State University. He would like it to be easier to provide seminars for his class, from industry professionals, to give them real world examples of the subject they are studying. This web application will make that process much easier for him and any professors looking to do the same.

2.2 Product Functions

The major functions of AI Connect include:

- *Seminar proposal submission by users.*
- *Seminar request submissions by industry professionals.*
- *Administrative functions to manage user accounts, seminars, and system data.*
- *User communication for coordinating and confirming seminars.*

2.3 User Classes and Characteristics

The two major user classes will be general, and admin. General users will be most frequent on the website with the ability to propose and request seminars. Admins will be managing these users and will be able to remove users to deal with malicious actors.

2.4 Operating Environment

This project will be a web application, so it should be available on any device with access to a web browser.

2.5 Design and Implementation Constraints

This application will be developed in English, though it may provide a barrier to some it is the most common language so it will be highly accessible. The application may also be difficult for those with disabilities to use, such as the visually impaired.

2.6 User Documentation

The following user documentation will be delivered alongside the software:

- *User manual covering basic and advanced system functionalities.*
- *Online help and tutorials accessible via the website.*

2.7 Assumptions and Dependencies

The application assumes access to reliable internet for most users. Dependencies include integration with third-party authentication systems (e.g., OAuth), and it may require the use of commercial web services for hosting.

3. External Interface Requirements

3.1 User Interfaces

The user interface will follow a modern, responsive design that ensures compatibility with both desktop and mobile browsers. It will include:

- *A dashboard for users to view seminars.*
- *Admin-specific panels to manage user accounts.*
- *Error handling messages and prompts.*

3.2 Hardware Interfaces

AI Connect interacts only with client-side hardware through web browsers. No specific hardware interfaces are required beyond standard network connectivity.

3.3 Software Interfaces

The application will interface with:

- *Web Servers: The system will interact with standard HTTP/HTTPS protocols.*
- *Database: The backend will connect to a relational database for storing seminar and user information.*

3.4 Communications Interfaces

The system will rely on HTTP and HTTPS protocols for communication, including security protocols for data encryption during user authentication and sensitive transactions.

4. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 User Registration and Authentication

4.1.1 Description and Priority

User should be able to register and account and the account should be authenticated.

4.1.2 Stimulus/Response Sequences

The user should be authenticated when they login.

4.1.3 Functional Requirements

- Users (universities, instructors, and industry professionals) should be able to log in using multiple methods, including email/password, LinkedIn, Google, and other OAuth providers.
- LinkedIn integration will enable users to import their profile data, including experience, skills, and professional history.
- Email/password registration should allow users to manually input their profile information.
- Password recovery functionality for email-based registrations.
- Secure OAuth for LinkedIn, Google, and other third-party logins.

4.2 Profile Management

4.2.1 Description and Priority

The user should be able to edit their profile.

4.2.2 Stimulus/Response Sequences

When the user clicks on their profile name they should be taken to an edit profile screen.

4.2.3 Functional Requirements

- Industry professionals can import their profile information directly from LinkedIn, while universities and instructors can manually enter their details.
- Users can update their profiles, adding or modifying their expertise, skills, institution details, and availability.
- Regular syncing for LinkedIn users to keep their profiles updated automatically.

4.3 Request and Matching System

4.3.1 Description and Priority

The user should be able to request and attend talks and be matched with similar skills and topics

4.3.2 Stimulus/Response Sequences

When the user requests a talk that request should be send to professionals looking for the same.

4.3.3 Functional Requirements

- Instructors, universities, and industry professionals can create requests to either teach or attend sessions, specifying topics, dates, and preferred delivery methods (online or in-person).
- Underserved institutions (schools, colleges, universities) can request voluntary teaching, specifying the subject and duration.
- The system automatically matches requests with relevant professionals based on their skills, topics, and availability.
- Notifications are sent when a relevant request or offer is made.

4.4 Text-Based Messaging for Requests

4.4.1 Description and Priority

The user should be able to message with someone they have matched and accepted a request from

4.4.2 Stimulus/Response Sequences

When a user accepts a request they should be put into a message with the requester

4.4.3 Functional Requirements

- Users should be able to text each other for coordination and clarification after a request is made.
- Real-time notifications for incoming texts and updates.
- Support for threaded conversations to manage request negotiations, confirmations, and clarifications.

4.5 Scheduling and Coordination

4.5.1 Description and Priority

The user should be provided with a calendar to help keep up with talks.

4.5.2 Stimulus/Response Sequences

The user should get notifications from their calendar for events.

4.5.3 Functional Requirements

- Users should manage their availability and schedule sessions via an integrated calendar.

- Automated reminders and notifications for upcoming sessions should be sent to both parties.
- Users can specify their session mode (online or in-person) and use messaging to coordinate logistics.

4.6 Remote Presentation Platform Integration

4.6.1 Description and Priority

Users can use third party platforms like zoom to make presentations.

4.6.2 Stimulus/Response Sequences

When scheduling a session users should be able to choose a platform.

4.6.3 Functional Requirements

- Integration with remote presentation platforms such as Zoom, Webex, and ON24 for online sessions.
- Users can choose the platform when scheduling a session, and the system will generate links and reminders.

4.7 Payments (Future Feature)

4.7.1 Description and Priority

The user should be able to pay for subscription models and other services.

4.7.2 Stimulus/Response Sequences

There should be integration with third party payment processors

4.7.3 Functional Requirements

- Integration with payment gateways (e.g., Stripe, PayPal) for charging users in the future.
- Support for generating invoices and receipts for completed transactions.
- Implementation of subscription models for premium users or transaction-based fees for services.
- Payments for voluntary teaching can be facilitated for honorariums.

4.8 Feedback, Ratings, and Talk History

4.8.1 Description and Priority

This feature allows users to provide feedback and ratings after each session. Additionally, the system maintains a history of sessions with feedback, ratings, and statistics, visible on each user's profile.

4.8.2 Stimulus/Response Sequences

After a session ends, the user submits feedback and a rating (e.g., 1-5 stars), prompting the system to store the feedback and update the user profile.

A user accesses their profile to view their session history, and the system displays the number of sessions given or attended, along with dates and associated feedback.

4.8.3 Functional Requirements

- *The system shall allow users to provide a rating (1-5 stars) and written feedback after each session.*
- *The system shall display a history of all previous sessions, including dates and feedback, on the user's profile.*
- *The system shall track the number of sessions a user has given or attended.*

4.9 Voluntary Teaching for Underserved Areas

4.9.1 Description and Priority

This feature supports voluntary teaching for schools, colleges, and universities in underserved areas. Educational institutions can request voluntary teaching, and professionals can offer their services for free, delivered remotely.

4.9.2 Stimulus/Response Sequences

An educational institution submits a request for voluntary teaching, which the system then displays to professionals signed up for voluntary teaching.

A professional selects a voluntary teaching opportunity, prompting the system to schedule the session and provide a link for remote delivery via Zoom or Webex.

4.9.3 Functional Requirements

- *The system shall allow educational institutions to request voluntary teaching.*
- *The system shall enable professionals to sign up for voluntary teaching and list their available times and subjects.*
- *The system shall support remote delivery of sessions through platforms like Zoom or Webex.*

4.10 Search and Filter Functionality

4.10.1 Description and Priority

This feature enables users to search for professionals or teaching requests based on topics, skills, or availability, facilitating easier matchmaking between educators and institutions.

4.10.2 Stimulus/Response Sequences

A user searches for professionals by topic or skill, resulting in the system displaying a list of relevant professionals matching the search criteria.

A professional searches for teaching requests from institutions, prompting the system to display a list of relevant requests based on availability, topic, or skill.

4.10.3 Functional Requirements

- *The system shall allow users to search for professionals based on topics, skills, and availability.*
- *The system shall allow professionals to search for teaching requests from educational institutions.*

4.11 Notifications and Alerts

4.11.1 Description and Priority

The user should be able to receive notifications and alerts.

4.11.2 Stimulus/Response Sequences

The user should receive notifications for any new interactions with configurable options for email, sms, and in platform notifications

4.11.3 Functional Requirements

- *Users should receive notifications for new requests, text messages, session reminders, and payment confirmations.*
- *Configurable notification settings for email, SMS, or in-platform notifications.*

4.12 Reporting and Analytics

4.12.1 Description and Priority

The user should have their stats tracked.

4.12.2 Stimulus/Response Sequences

The users dashboard should have their tracked stats.

4.12.3 Functional Requirements

- *Each user should have a dashboard displaying session history, feedback received, requests made and responded to, and talks delivered.*
- *Admins should have access to analytics for monitoring platform usage, session counts, and user engagement.*
- *The system should store records of all sessions (upcoming, in-progress, completed) for tracking and reporting purposes.*

4.13 Security and Privacy

4.13.1 Description and Priority

The users data should be protected.

4.13.2 Stimulus/Response Sequences

A user should be authenticated when the login to the system.

4.13.3 Functional Requirements

- Ensure secure storage of user data, with encryption for personal and professional information.
- Users can control what data is visible to others on their profiles.
- Role-based access control (RBAC) to limit access to sensitive data.
- Ensure compliance with regulations like GDPR and CCPA.

4.14 Admin Functionality

4.14.1 Description and Priority

The admin should be able to manage users and session requests.

4.14.2 Stimulus/Response Sequences

The admin should remove and malicious users or any malicious session requests.

4.14.3 Functional Requirements

- Admins should have the ability to manage all users, including the ability to view, block, or suspend users as needed.
- Admins can view user statistics, such as the number of sessions attended, talks given, and requests made.
- Admins can moderate feedback, handle disputes, and ensure compliance with platform rules.
- Reporting tools should allow admins to monitor platform activity and take action on violations.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The platform should handle up to 1000 concurrent users with minimal response times under 3 seconds for loading main pages. The backend must process seminar submissions within 2 seconds.

5.2 Safety Requirements

There are no specific safety concerns for AI Connect, but the platform should ensure that all user data is secure.

5.3 Security Requirements

User authentication is required for all system functions and all data transmissions will be encrypted.

5.4 Software Quality Attributes

- *Usability: The platform should be intuitive, with minimal learning curve for professors and industry professionals.*
- *Reliability: The system should have 99.9% uptime to meet academic schedules.*
- *Maintainability: Code should be modular to allow easy future updates.*

5.5 Business Rules

Admins are responsible for approving seminar proposals and handling user management. Professors have final say on which seminars are approved.

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

AI Connect: The name of the web application.